Amazon Elastic Load Balancer (ELB) — Complete Interview Questions & Answers

1. What is Elastic Load Balancer?
Answer:
Elastic Load Balancer, or ELB, is an AWS service used to distribute incoming traffic across multiple servers or application resources. It helps prevent one server from getting all the traffic. ELB also improves application availability and can automatically send traffic only to healthy targets.
For example, suppose we have an application running on three EC2 instances. Instead of users directly connecting to one EC2 instance, they connect to the Load Balancer. The Load Balancer receives the request and sends it to one of the healthy EC2 instances.
Example:
User
↓
Load Balancer
↓

↓ ↓ ↓
EC2-1 EC2-2 EC2-3
Simple ga gurthupettuko:
ELB = Distribute traffic + High availability + Healthy servers

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2. Why do we need a Load Balancer?
Answer:
We use a Load Balancer when we have multiple servers running the same application. It distributes incoming traffic between those servers instead of sending all requests to one server. This helps improve performance, availability, and reliability.
Example:
Without Load Balancer:
Users
↓
EC2-1
If EC2-1 goes down, the application becomes unavailable.
With Load Balancer:
Users
↓
Load Balancer
↓

↓ ↓ ↓
EC2-1 EC2-2 EC2-3
If EC2-1 goes down, the Load Balancer can send traffic to EC2-2 or EC2-3.

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3. How does a Load Balancer work?
Answer:
A Load Balancer receives requests from users and checks the available targets behind it. It then chooses a suitable healthy target and forwards the request to that target. The response comes back through the Load Balancer to the user.
Example:
User
↓
Load Balancer
↓
Check available targets
↓
Choose healthy target
↓
EC2
↓
Response
↓
User

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4. What are the main types of Elastic Load Balancers?
Answer:
AWS provides different types of Load Balancers for different types of traffic. The main types are Application Load Balancer, Network Load Balancer, Gateway Load Balancer, and Classic Load Balancer. Classic Load Balancer is the older generation and is mainly seen in old applications.
Main types:
ELB
│
├— Application Load Balancer (ALB)
├— Network Load Balancer (NLB)

├── Gateway Load Balancer (GWLB)
└── Classic Load Balancer (CLB)

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5. What is an Application Load Balancer?
Answer:
Application Load Balancer, or ALB, works at the application level and is mainly used for HTTP and HTTPS traffic. It can make routing decisions based on things like host name, URL path, HTTP headers, and query parameters. It is commonly used for web applications and APIs.
Example:
www.myapp.com
↓
ALB
↓
EC2 Application
For example, /users can go to one target group and /orders can go to another target group.

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6. What is a Network Load Balancer?
Answer:
Network Load Balancer, or NLB, is mainly used for TCP, TLS, and UDP traffic. It works at the network level and is designed for high performance and very low latency. It is useful when applications need fast network connections and large amounts of traffic.
Example:
Client
↓
NLB
↓

↓ ↓
Server-1 Server-2
Simple ga gurthupettuko:
NLB = Network traffic + High performance

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7. What is a Gateway Load Balancer?
Answer:
Gateway Load Balancer, or GWLB, is mainly used to deploy and scale network security appliances. It can send traffic through devices such as firewalls, intrusion detection systems, and other network security tools. It is useful when network traffic needs to pass through security appliances.
Example:
User Traffic
↓
Gateway Load Balancer
↓
Firewall / Security Appliance
↓
Application

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8. What is a Classic Load Balancer?
Answer:
Classic Load Balancer, or CLB, is the older type of AWS Load Balancer. It supports basic HTTP, HTTPS, and TCP load balancing. For new applications, AWS generally recommends using the newer load balancer types based on the application requirement.
Example:
Users
↓
Classic Load Balancer
↓
EC2 Instances

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9. What is the difference between ALB and NLB?
Answer:
ALB is mainly used for HTTP and HTTPS applications and can make routing decisions using application-level information such as URL paths and host names. NLB is mainly used for TCP, TLS, and UDP traffic and is designed for high performance and low latency.
Example:
ALB
↓
HTTP / HTTPS
↓
Web Application
NLB
↓
TCP / TLS / UDP
↓
Network Application

Simple ga gurthupettuko:
ALB → Application traffic
NLB → Network traffic

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10. What is a Load Balancer Listener?
Answer:
A Listener is a process that checks for incoming connection requests on a specific protocol and port. For example, an ALB can have a listener for HTTP on port 80 and HTTPS on port 443. The listener then uses its rules to decide where the request should go.
Example:
User
↓
ALB
↓
Listener : 443 HTTPS
↓
Listener Rule
↓
Target Group

---

11. What is a Listener Rule?
Answer:
A Listener Rule tells the Load Balancer what to do when a request matches a particular condition. In an ALB, rules can use information such as host name or URL path. Based on the rule, the request can be sent to a particular target group.
Example:
www.myapp.com/users
↓
ALB
↓
Path = /users
↓
Users Target Group
Another request:
www.myapp.com/orders
↓
ALB
↓
Path = /orders
↓
Orders Target Group

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12. What is a Target in Load Balancing?
Answer:
A target is the resource that receives traffic from the Load Balancer. Depending on the Load Balancer and configuration, targets can include EC2 instances, IP addresses, containers, or other supported resources.
Example:
ALB
↓
Target Group
↓

↓ ↓ ↓
EC2-1 EC2-2 EC2-3
Here, EC2-1, EC2-2, and EC2-3 are targets.

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13. What is a Target Group?
Answer:
A Target Group is a collection of targets that a Load Balancer sends traffic to. It also contains settings such as the health check configuration. A listener forwards requests to a target group based on its rules.
Example:
ALB
↓
Listener
↓
Target Group
↓

↓ ↓ ↓
EC2-1 EC2-2 EC2-3

---

14. Why do we need Target Groups?
Answer:

Target Groups help the Load Balancer organize and manage the servers that should receive traffic. We can create different target groups for different applications or services. Listener rules can then send requests to the correct target group.

Example:

ALB
│
├── /users
│ ↓
│ Users Target Group
│ ↓
│ EC2-1, EC2-2
│
└── /orders
↓
Orders Target Group
↓
EC2-3, EC2-4

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15. What is a Health Check?

Answer:

A Health Check is used by the Load Balancer to check whether a target is working correctly. The Load Balancer regularly sends a request to the target and checks the response. If the target is unhealthy, the Load Balancer stops sending normal traffic to it.

Example:

Load Balancer
↓
Health Check
↓

↓ ↓
Healthy Unhealthy
EC2-1 EC2-2
↓
Receives No normal
traffic traffic

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16. Why are Health Checks important?

Answer:

Health Checks prevent the Load Balancer from sending traffic to a failed or unhealthy server. This helps improve application availability and gives users a better experience. When the target becomes healthy again, the Load Balancer can send traffic to it again.

Example:

ALB
↓
Health Check
↓
EC2-1 → Healthy → Send traffic
EC2-2 → Unhealthy → Stop traffic
EC2-3 → Healthy → Send traffic

---

17. What happens if one EC2 instance goes down?

Answer:

If a target fails its health checks, the Load Balancer marks it as unhealthy and stops sending normal traffic to that target. Traffic is then sent to other healthy targets in the target group. When the failed target becomes healthy again, it can receive traffic again.

Example:

ALB
↓

↓ ↓ ↓
EC2-1 EC2-2 EC2-3
Healthy Failed Healthy
↓ ↓
Traffic Traffic

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18. What is Cross-Zone Load Balancing?

Answer:

Cross-Zone Load Balancing allows a Load Balancer to distribute traffic across targets in different Availability Zones. This helps distribute traffic more evenly when targets are not equally distributed across the Availability Zones.

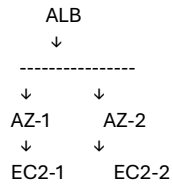
Example:

ALB
│
├── AZ-1 → EC2-1
│

└── AZ-2 → EC2-2, EC2-3
The Load Balancer can distribute traffic across the available healthy targets.

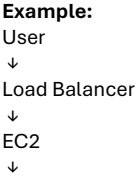
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19. What is High Availability in Load Balancing?
Answer:
High Availability means keeping the application available even if one server or Availability Zone has a problem. We normally place application targets in multiple Availability Zones and use a Load Balancer in front of them. If one target becomes unavailable, traffic can go to healthy targets.
Example:



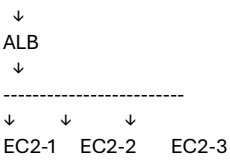
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20. What is the difference between Load Balancer and EC2?
Answer:
EC2 is a compute service where our application runs, while a Load Balancer receives incoming traffic and distributes it to application servers. The Load Balancer does not replace EC2; it works in front of EC2 or other targets.
Example:



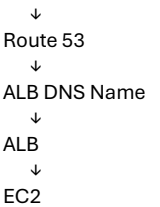
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21. Can a Load Balancer work with multiple EC2 instances?
Answer:
Yes. A Load Balancer is commonly used with multiple EC2 instances running the same application. It distributes incoming requests across the healthy instances. This helps us handle more traffic and improves availability.
Example:



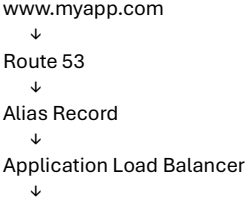
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22. What is DNS name of a Load Balancer?
Answer:
When we create a Load Balancer, AWS provides a DNS name for it. Users or other services can use this DNS name to reach the Load Balancer. In real applications, we commonly use our own domain name with Route 53 and point it to the Load Balancer.
Example:



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23. Can we connect Route 53 with an Application Load Balancer?
Answer:
Yes. Route 53 can be used to connect a domain name to an Application Load Balancer. We commonly create an Alias record in Route 53 that points the domain to the ALB. When users enter the domain, Route 53 sends them to the Load Balancer.
Example:



EC2 Application

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24. What is an ALB Listener Port?
Answer:
A Listener Port is the port on which the Load Balancer receives incoming traffic. Common examples are port 80 for HTTP and port 443 for HTTPS. The listener then applies its rules and forwards the request to the correct target group.
Example:
User
↓
ALB
↓
Port 443
↓
HTTPS Listener
↓
Target Group

---

25. What is HTTP and HTTPS in Load Balancing?
Answer:
HTTP is used for normal web traffic, while HTTPS is the secure version of HTTP. In production applications, HTTPS is commonly used to protect data between the user and the Load Balancer. An ALB can have listeners for both HTTP and HTTPS.
Example:
User
↓
HTTPS : 443
↓
ALB
↓
EC2

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26. What is SSL/TLS Termination?
Answer:
SSL/TLS termination means the HTTPS connection from the user is decrypted at the Load Balancer. The Load Balancer can then forward the request to the backend target using HTTP or HTTPS based on the configuration. This removes some HTTPS processing work from the backend servers.
Example:
User
↓
HTTPS
↓
ALB
↓
HTTP / HTTPS
↓
EC2

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27. What is an SSL Certificate in ALB?
Answer:
An SSL certificate is used to enable HTTPS on the Load Balancer. AWS Certificate Manager, or ACM, can provide and manage certificates that can be used with supported AWS services such as ALB. The certificate helps secure communication between the user and the Load Balancer.
Example:
User
↓
HTTPS
↓
ALB
↓
SSL Certificate
↓
Application

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28. What is AWS Certificate Manager?
Answer:
AWS Certificate Manager, or ACM, is an AWS service used to create, manage, and deploy SSL/TLS certificates. We can use an ACM certificate with an Application Load Balancer to enable HTTPS. This helps secure traffic between users and the Load Balancer.
Example:
ACM
↓
SSL Certificate
↓

ALB HTTPS Listener
↓
Application

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29. What is Path-Based Routing?
Answer:
Path-Based Routing means routing traffic based on the URL path in the request. An ALB can send different paths to different target groups. This is useful when multiple services are running behind one Load Balancer.
Example:
www.myapp.com/users
↓
ALB
↓
Users Target Group
www.myapp.com/orders
↓
ALB
↓
Orders Target Group

---

30. What is Host-Based Routing?
Answer:
Host-Based Routing means routing traffic based on the domain name used by the user. An ALB can send different domain names to different target groups. This allows multiple applications to use the same Load Balancer.
Example:
users.myapp.com
↓
ALB
↓
Users Target Group
orders.myapp.com
↓
ALB
↓
Orders Target Group

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31. What is Weighted Routing in ALB?
Answer:
Weighted routing allows us to send different amounts of traffic to different target groups. For example, we can send 90% of traffic to the old application and 10% to the new application. This can be useful for testing or gradual application releases.
Example:
ALB
│
├── 90% → Old Application
│
└── 10% → New Application

---

32. What is Sticky Session?
Answer:
Sticky Session means requests from the same user can continue going to the same target for a period of time. This is useful for applications that temporarily keep user information on a particular server. ALB supports session stickiness through supported mechanisms.
Example:
User A
↓
ALB
↓
EC2-1

Next request from User A
↓
ALB
↓
EC2-1

---

33. What is the problem with Sticky Sessions?
Answer:
Sticky Sessions can make traffic distribution less even because a user may continue going to the same server. If that server receives many sticky users, it can become more loaded than others. For this reason, applications are often designed to keep session data outside individual servers.
Example:

Users
↓
ALB
↓
EC2-1 ← Many sticky users
EC2-2
EC2-3

---

34. What is Connection Draining?
Answer:
Connection Draining means allowing existing connections to finish before a target is removed from service. This helps prevent active users from suddenly losing their connection when a server is being removed or replaced. In newer AWS terminology, this behavior is generally handled through target deregistration delay.
Example:
EC2-1
↓
Being removed
↓
Existing requests finish
↓
No new traffic
↓
EC2-1 removed

---

35. What is Deregistration Delay?
Answer:
Deregistration Delay defines how long the Load Balancer waits for existing connections to finish before fully removing a target from service. During this period, new requests are not normally sent to the deregistering target. It helps application users complete active requests without sudden interruption.
Example:
EC2
↓
Deregister
↓
Wait for existing requests
↓
Finish connections
↓
Remove target

---

36. What is Auto Scaling with Load Balancer?
Answer:
Auto Scaling can automatically increase or decrease the number of EC2 instances based on application demand. The Load Balancer distributes traffic across the available instances. Together, Auto Scaling and the Load Balancer help applications handle changing traffic.
Example:
High Traffic
↓
Auto Scaling
↓
More EC2 Instances
↓
Load Balancer
↓
Traffic Distributed

---

37. How does Load Balancer work with Auto Scaling Group?
Answer:
We can attach a target group to an Auto Scaling Group. When Auto Scaling launches a new EC2 instance, it can automatically register that instance with the target group. When an instance is removed, it can also be removed from the target group.
Example:
Auto Scaling Group
↓

↓ ↓ ↓
EC2-1 EC2-2 EC2-3
↓
Target Group
↓
ALB

---

38. What happens when Auto Scaling launches a new EC2 instance?
Answer:
When a new EC2 instance is launched by an Auto Scaling Group, it can automatically be registered with the Load Balancer's target group. The Load Balancer performs health checks on the new instance. Once it becomes healthy, it can start receiving traffic.
Example:
Auto Scaling
↓
New EC2
↓
Target Group
↓
Health Check
↓
Healthy
↓
Receives Traffic

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39. What is Load Balancer Security Group?
Answer:
A Load Balancer Security Group controls which traffic is allowed to reach the Load Balancer. For example, we can allow users from the internet to reach the ALB on ports 80 and 443. We should allow only the required traffic.
Example:
Internet
↓
ALB Security Group
↓
Allow 80 / 443
↓
ALB

---

40. What is the Security Group of EC2 behind ALB?
Answer:
The EC2 Security Group should normally allow traffic from the Load Balancer Security Group rather than allowing everyone from the internet. This creates a safer network design where users reach the application through the Load Balancer.
Example:
Internet
↓
ALB Security Group
↓
ALB
↓
EC2 Security Group
↓
EC2

---

41. Why should we not directly allow internet traffic to EC2?
Answer:
If users can directly access the EC2 instance, they can bypass the Load Balancer. This can make it harder to control traffic and can reduce the security of the application. A common design is to allow users to access the ALB and allow the EC2 instances to receive traffic only from the ALB.
Example:
Internet
↓
ALB
↓
EC2
Instead of:
Internet
↓
EC2 directly

---

42. What is the difference between Public and Internal Load Balancer?
Answer:
An internet-facing Load Balancer can receive traffic from the internet and is commonly used for public applications. An internal Load Balancer is used inside a VPC and is mainly used for private applications or communication between internal services.
Example:
Public:
Internet
↓
Internet-facing ALB
↓

Application
Internal:
Service A
↓
Internal ALB
↓
Service B

---

43. What is an Internal Load Balancer?
Answer:
An Internal Load Balancer has private IP addresses and is reachable only from the allowed private network. It is useful when we do not want the application to be directly available from the internet. It is commonly used for internal services and microservices.
Example:
Frontend
↓
Internal ALB
↓
Backend Service

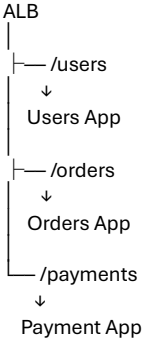
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44. What is an Internet-facing Load Balancer?
Answer:
An Internet-facing Load Balancer can receive requests from users over the internet. It is commonly used when we have a public web application. It sends the incoming requests to healthy backend targets.
Example:
Internet
↓
Internet-facing ALB
↓
EC2

---

45. What is the difference between Internet-facing and Internal ALB?
Answer:
An Internet-facing ALB is designed to receive traffic from the internet, while an Internal ALB is designed for private network traffic. Both can distribute traffic to backend targets, but they are used for different network access requirements.
Example:
Internet
↓
Internet-facing ALB
↓
Frontend
Frontend
↓
Internal ALB
↓
Backend

---

46. Can ALB route traffic to different applications?
Answer:
Yes. An ALB can use listener rules to route requests to different target groups. We can route based on host name, URL path, and other supported request conditions.
Example:
ALB


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47. Can ALB route traffic based on domain name?
Answer:
Yes. ALB supports host-based routing, where the request is routed based on the domain name. This allows multiple applications or services to share one Load Balancer.
Example:
users.myapp.com

↓
ALB
↓
Users Application
payments.myapp.com
↓
ALB
↓
Payment Application

---

48. What is the difference between Listener and Target Group?
Answer:
A Listener receives incoming traffic on a specific port and protocol and checks the listener rules. A Target Group contains the backend targets that can receive traffic. The listener forwards requests to a target group based on its rules.
Example:
User
↓
Listener
↓
Target Group
↓
EC2
Simple ga gurthupettuko:
Listener → Receives traffic
Target Group → Contains backend targets

---

49. What is the difference between Load Balancer and Target Group?
Answer:
The Load Balancer is the main entry point that receives incoming traffic. A Target Group is a group of backend resources that can receive traffic from the Load Balancer. The Load Balancer uses listeners and rules to decide which target group should receive a request.
Example:
User
↓
Load Balancer
↓
Target Group
↓
EC2-1
EC2-2
EC2-3

---

50. What happens if all targets become unhealthy?
Answer:
If all targets fail their health checks, the Load Balancer has no healthy target to send normal traffic to. Users may then receive errors or unavailable responses depending on the configuration. This is why we need healthy backend instances and proper monitoring.
Example:
ALB
↓

↓ ↓ ↓
EC2-1 EC2-2 EC2-3
✕ ✕ ✕
↓
No healthy target

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51. What is Load Balancer Access Log?
Answer:
Access Logs contain information about requests received by the Load Balancer. They can help us understand who made requests, which URL was requested, when the request happened, and how the request was handled. They are useful for troubleshooting and traffic analysis.
Example:
User Request
↓
ALB
↓
Access Log
↓
Request Information

---

52. Why do we use Load Balancer logs?

Answer:
Load Balancer logs are useful when we need to investigate traffic or application problems. They can help us understand request patterns, response status codes, client information, and timing information. This makes troubleshooting easier.
Example:
User
↓
ALB
↓
Request
↓
Access Log
↓
Troubleshooting

---

53. What is a Load Balancer Attribute?
Answer:
Load Balancer attributes are settings that control certain behaviors of the Load Balancer. Depending on the Load Balancer type, attributes can control features such as access logs, deletion protection, idle timeout, and other supported settings.
Example:
Load Balancer
↓
Attributes
↓
Settings

---

54. What is Deletion Protection in Load Balancer?
Answer:
Deletion Protection helps prevent an important Load Balancer from being accidentally deleted. When it is enabled, the Load Balancer cannot be deleted until deletion protection is disabled. It is useful for production environments.
Example:
Production ALB
↓
Deletion Protection = ON
↓
Accidental Delete
↓
Blocked

---

55. What is Idle Timeout?
Answer:
Idle Timeout defines how long the Load Balancer keeps an inactive connection open when no data is being sent. If there is no activity for the configured time, the connection can be closed. The value should be suitable for the application.
Example:
Client
↓
ALB
↓
Connection
↓
No activity
↓
Idle Timeout
↓
Connection closed

---

56. What is the Load Balancer Scheme?
Answer:
The Load Balancer scheme tells us whether the Load Balancer is internet-facing or internal. An internet-facing Load Balancer is reachable from the internet, while an internal Load Balancer is used inside private networks.
Example:
Scheme
│
├── Internet-facing
│
└── Internal

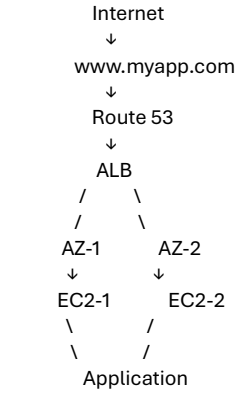
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57. What is the complete ALB request flow?
Answer:
When a user accesses the application, the request first reaches the domain and then the Load Balancer. The ALB listener receives the request, checks its rules, and selects the correct target group. The request is then sent to a healthy backend target.
Example:

User
↓
www.myapp.com
↓
Route 53
↓
ALB
↓
Listener
↓
Listener Rule
↓
Target Group
↓
Health Check
↓
Healthy EC2
↓
Application

---

58. Can you explain a real-world ALB architecture?
Answer:
In a common production setup, users access the application through a domain name. Route 53 sends the request to an internet-facing ALB, and the ALB distributes traffic to healthy EC2 instances running in multiple Availability Zones. Security Groups control access between the Load Balancer and the EC2 instances.
Example:



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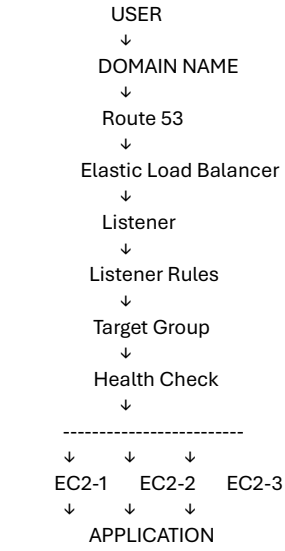
59. If the interviewer asks, "Explain Elastic Load Balancer in short," what should I say?
Answer:
“Elastic Load Balancer is an AWS service used to distribute incoming application traffic across multiple healthy targets. It helps improve application availability and prevents a single server from handling all the traffic. AWS provides different Load Balancer types such as ALB, NLB, and GWLB, depending on the traffic and application requirement. With ALB, we can also use listener rules, target groups, health checks, and different routing options.”
Example:
Users
↓
Elastic Load Balancer
↓

↓ ↓ ↓
EC2-1 EC2-2 EC2-3

---

60. How do you remember the complete Elastic Load Balancer flow?
Answer:
I remember ELB as a complete traffic flow from the user to the backend application. The user reaches the Load Balancer, the listener receives the request, the listener rule decides where to send it, and the target group contains the backend targets. Health checks make sure that traffic goes only to healthy targets.

Example:



Final ELB Memory Map

```
ELB
│
├── Load Balancer Types
│   ├── ALB
│   ├── NLB
│   ├── GWLB
│   └── CLB
│
├── ALB
│   ├── Listener
│   ├── Listener Rules
│   ├── Target Group
│   ├── Health Check
│   ├── Path-Based Routing
│   ├── Host-Based Routing
│   └── Sticky Sessions
│
├── Networking
│   ├── VPC
│   ├── Subnets
│   ├── Security Group
│   ├── Internet-facing
│   └── Internal
│
├── Availability
│   ├── Multiple AZs
│   ├── Healthy Targets
│   └── Cross-Zone Load Balancing
│
├── Security
│   ├── HTTPS
│   ├── SSL/TLS
│   └── ACM Certificate
│
├── Scaling
│   └── Auto Scaling Group
│
└── Monitoring / Troubleshooting
    ├── Health Checks
    ├── Access Logs
    └── CloudWatch
```

🔥 **Super-short interview memory**

ELB → Distribute Traffic

ALB → HTTP / HTTPS
NLB → TCP / TLS / UDP
GWLB → Security Appliances
Listener → Receives Traffic
Listener Rule → Decides Where Traffic Goes
Target Group → Group of Backend Targets
Health Check → Checks Target Health
ALB Routing → Path / Host Based
Sticky Session → Same User → Same Target
Multi-AZ → High Availability
Auto Scaling → Add / Remove EC2
ACM → SSL/TLS Certificate
Internal ALB → Private Traffic
Internet ALB → Internet Traffic
Access Logs → Request Information

One final line to remember:

ELB = Receive traffic → Check rules → Choose target group → Send traffic to healthy backend servers.